

$$\mathbf{e}_{t,k}^Z = P_Z \mathbf{W}_{t,k}^Z \left(\mathbf{Y}_{t,k}^Z - \hat{\mathbf{Z}}_{t,k}^1 \right),$$

$$v_{\text{FBFM}}^Z = v_{\theta_Z}^Z + \lambda_{\tau_k^Z}^Z \left(\mathbf{J}_{t,k}^Z \right)^\top \mathbf{e}_{t,k}^Z$$